Cost and Budget Management

Cost management estimates, budgets and controls the money the project needs, and connects effort to cost so that decisions can be made in currency.

PMBOK Guide 7th ed. | Earned value: ANSI/EIA-748 concepts | LibreOffice Calc · ProjectLibre

AGENDA

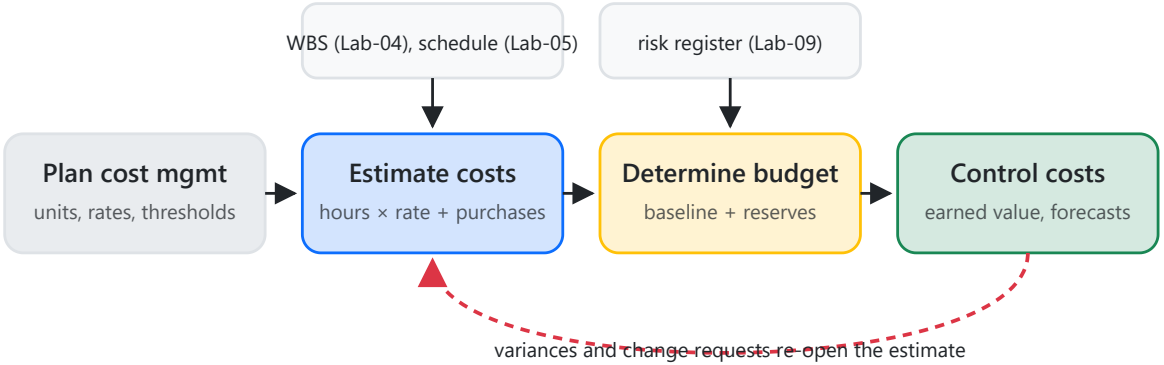
What we will cover

1. Cost types: direct, indirect, fixed, variable, sunk, opportunity
2. Estimating techniques: bottom-up, analogous, parametric, function points, story points to cost
3. Estimate accuracy ranges: rough order of magnitude to definitive
4. Cost of IT projects: people, licences, cloud and hosting, hardware, training, maintenance
5. Cost baseline, contingency and management reserves
6. Budget versus funding and cash flow over time
7. Earned value management: PV, EV, AC, SV, CV, SPI, CPI
8. Forecasting: EAC, ETC, TCPI and what to do when they slip
9. Total cost of ownership and return on investment for IT decisions
10. Cost control and change requests
11. Chapter Quiz
12. Lab-06

Click any item to jump directly to that topic.

From hours to money: the four cost processes

- Chapter 05 gave every activity a **duration** and a date. This chapter multiplies effort by a **rate**, adds the things you buy, and turns the plan into a **budget** the sponsor can approve.
- PMBOK Guide 7th ed. (Process Groups Practice Guide) names four processes: **plan** cost management, **estimate** costs, **determine budget**, **control** costs.
- Money is the common unit: a slipped week, a rented server and a bought licence can all be compared once they are in USD.
- Case study numbers used all chapter: 4 students at a notional **5 USD/h**, lecturer **20 USD/h**, Linode 4 GB VPS **25 USD/month**, domain **15 USD/year**.



output of this chapter: a cost baseline in USD that Lab-06 time-phases by week and tracks with EVM

TOPIC 1

Cost types: two questions about every euro or dollar

	Fixed (does not change with volume of work)	Variable (grows with hours, users or months)
Direct charged to this project only	Domain `gicitc.com` 15 USD/year · one-off SSL setup · a bought icon pack 45 USD	Student hours 576 h × 5 USD · lecturer 28 h × 20 USD · VPS 25 USD × months the project runs
Indirect shared with other projects, allocated	Lab room, campus Wi-Fi, GitHub organisation, the lecturer's salary base	Electricity of the lab, shared CI minutes, a departmental Jira licence billed per seat

Also	Meaning	Realtime Quiz example
Sunk	Already spent, cannot be recovered; must **not** influence the decision to continue	The 1,200 USD of hours used by week 6 when the team asks "should we switch to a SaaS tool?"
Opportunity	Value of the best alternative you give up	576 student hours could have built the mobile client instead

- **Direct or indirect**: can the invoice or timesheet be attributed to this project alone? Indirect costs enter the budget as an *overhead rate* (for example +15 % on labour) or are left out and stated as an assumption.
- **Fixed or variable**: does the amount move when the project grows by a week or by 100 users? Variable costs are the ones that estimating errors multiply.
- **Sunk** costs are history. The rational question is always "what does it cost *from here* to finish, and what is it worth?" (PMBOK Guide 7th ed., Measurement performance domain).
- **Opportunity cost** has no invoice but decides trade-offs: the sponsor compares projects by what else the same people could deliver.

In the labs the students' hours are *notional* (nobody is paid), but costing them at 5 USD/h makes effort and cash comparable in one number.

Worked example: "we already spent 1,200 USD, we cannot stop now"

```
Situation, end of week 6: a free SaaS quiz tool appears that covers
the MVP. Spent so far on Realtime Quiz: 1,200 USD (hours at 5 USD/h).

Option A  continue building        Option B  switch to SaaS
  cost from here to go-live          cost from here to go-live
  remaining hours 440 h × 5 = 2,200    migration of 3 quizzes  8 h ×  5 =   40
  hosting 2 months  2 × 25  =    50    lecturer training       4 h × 20 =   80
  --------------------------------    SaaS fee to week 14   2 × 17 × 10 = 340
  cost to finish            = 2,250   --------------------------------
  benefit: own data, exports,         cost to finish             =   460
  learning outcome of the capstone    benefit: MVP only, no CSV/PDF export

Wrong reasoning: "A has cost 1,200 + 2,250 = 3,450, B only 460 + 1,200 wasted".
Right reasoning: the 1,200 is spent in BOTH options; compare 2,250 vs 460
                 against the value of the extra benefits (and the grade).
```

- Write the two columns **from today forward**. The past sits above the line in both, so it cancels out.
- Put non-financial benefits in words under each column. The decision is "is the extra benefit of A worth 2,250 − 460 = 1,790 USD?", not "how do we save the 1,200".
- Opportunity cost appears here too: the 440 remaining hours are the price of the learning outcome. For a capstone that outcome is the point, so A wins; for a department that only needs quizzes, B wins.

⚠ "Escalation of commitment" is the classic failure: teams keep funding a project because of what it already cost. A cost management plan should name the *decision points* where the from-here comparison is redone (for Realtime Quiz: the MVP checkpoint in week 7).

Estimating techniques: four ways to get to a number

Technique	How	Accuracy · cost to produce	Realtime Quiz
Analogous (top-down)	Scale a similar past project	low · minutes	Last year's "Campus Library" web app took 500 h; this one has more real-time work: 500 × 1.2 ≈ 600 h
Parametric	Size unit × historical rate per unit	medium · hours, needs data	120 story points × 6 h/point = 720 h
Three-point (PERT)	(O + 4M + P) / 6 per element	medium, gives a range	Quiz authoring: O 80, M 96, P 160 → 104 h
Bottom-up	Estimate each work package, roll up the WBS	high · days, needs a WBS	10 branches, 576 student hours + 28 lecturer hours + purchases = 3,600 USD

Function points (IFPUG) count inputs, outputs, inquiries, files and interfaces; the industry converts them with a productivity figure (for example 8 h per FP for a Java web team). Story points are the agile relative-size unit; they become cost only through the team's *measured* hours per point.

- Use the cheap techniques **early** (charter, business case) and the expensive one **once the WBS exists**. Then compare: if parametric and bottom-up differ by more than 20 %, one of them hides a wrong assumption.
- Every estimate needs its **basis of estimate**: assumptions (rate, productivity, exclusions), source data and the accuracy range. An unexplained number is a guess.
- PMBOK Guide 7th ed. lists these under the Planning performance domain; the Agile Practice Guide adds relative sizing (story points, T-shirt sizes).

💡 Analogous 600 h, parametric 720 h, bottom-up 604 h (576 + 28): three independent routes landing within 20 % is the best evidence a student estimate can offer.

Parametric from story points, three-point for one branch

```
PARAMETRIC (whole product)
  backlog size          120 story points (product backlog, Lab-03)
  historical rate       6 h per point  (Software Engineering course
                        teams, last two semesters: 5.4 to 6.8 h/pt)
  effort  = 120 pt × 6 h/pt          = 720 h
  cost    = 720 h × 5 USD/h          = 3,600 USD  (labour only)
  range   = 120 × 5.4 .. 120 × 6.8   = 648 .. 816 h
          = 3,240 .. 4,080 USD

  Basis of estimate: points from the refined backlog; rate from
  two cohorts; excludes lecturer time, hosting and tools.
```

```
THREE-POINT (WBS 4 Quiz authoring, beta / PERT distribution)
  O  optimistic   80 h   (entities + API reuse Lab-03 code)
  M  most likely  96 h   (bottom-up sum of five activities)
  P  pessimistic 160 h   (validation rules keep changing)

  E  = (O + 4M + P) / 6 = (80 + 384 + 160) / 6 = 624 / 6 = 104 h
  SD = (P - O) / 6      = 80 / 6                       = 13.3 h

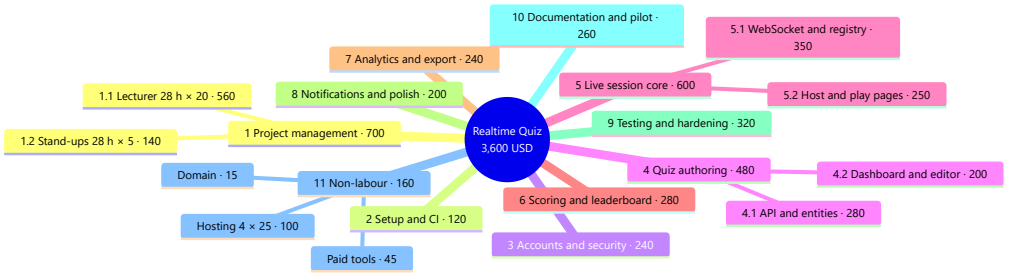
  cost E     = 104 h × 5 USD  = 520 USD
  cost ±1 SD = (104 ± 13.3) × 5 = 453 .. 587 USD (68 % confidence)
  cost ±2 SD = (104 ± 26.7) × 5 = 387 .. 653 USD (95 % confidence)

  The bottom-up M gives 480 USD; the 40 USD gap is the first
  candidate for contingency (slide 13).
```

💡 Parametric models are only as good as the **rate**. Record hours per point every sprint (Lab-05 velocity) so next semester's rate is yours, not borrowed.

⚠ Three-point in a spreadsheet: `=(B2+4*C2+D2)/6` for E, `=(D2-B2)/6` for SD. Never add the pessimistic values of every branch: that is not a range, it is a fantasy budget.

Bottom-up: estimate the leaves, roll up the WBS



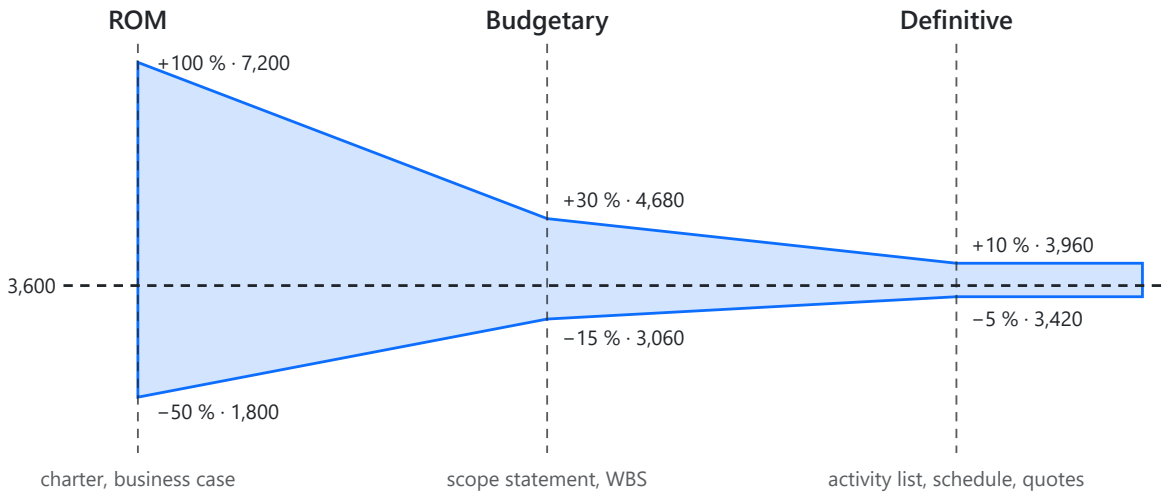
estimate.csv, branch 4 (rate in D, roll-up formulas on row 7)

	A	B	C	D	E	F	G
1	ID	Activity	Hrs	Rate	Labour	Other	Total
2	4.1.1	Quiz/Question entities	16	5	=C2*D2	0	=E2+F2
3	4.1.2	REST API + validation	24	5	=C3*D3	0	=E3+F3
4	4.1.3	API tests	16	5	=C4*D4	0	=E4+F4
5	4.2.1	Instructor dashboard	20	5	=C5*D5	0	=E5+F5
6	4.2.2	Quiz editor page	20	5	=C6*D6	0	=E6+F6
7	4	Quiz authoring	=SUM(C2:C6)		=SUM(E2:E6) ...		=SUM(G2:G6)

row 7 shows: 96 h, 480 USD

- Leaves are the **activities** of Lab-05; each gets hours × rate plus purchases.
- Parents are **formulas**, never typed numbers, so a changed leaf updates the total.
- Students 576 h × 5 = 2,880; lecturer 560; non-labour 160; total **3,600 USD**.

Accuracy ranges: the funnel narrows as you learn



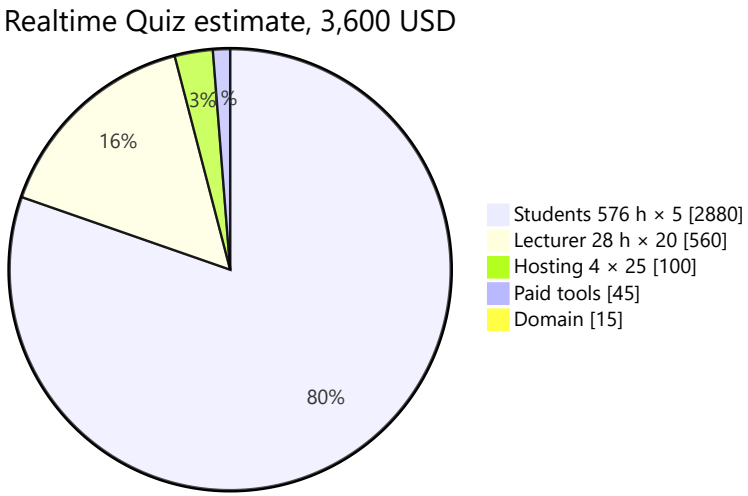
life cycle → the range is a property of the information available, not of the estimator's skill

Class	Range	Based on	3,600 USD becomes
Rough order of magnitude	−50 % / +100 %	analogy, expert judgement	1,800 to 7,200
Budgetary (preliminary)	−15 % / +30 %	parametric, WBS level 2	3,060 to 4,680
Definitive	−5 % / +10 %	bottom-up on activities, quotes	3,420 to 3,960

- Ranges are asymmetric: things go wrong more often than they go unexpectedly well.
- Always state the class with the number: "3,600 USD, definitive, −5/+10 %". PMBOK Guide 7th ed. quotes −25/+75 % as a typical ROM; AACE class 5 uses −50/+100 %. Agree the convention in the cost management plan.

⚠ The sponsor remembers the first number. If you say "about 2,000" in week 0, write "ROM, could be 4,000" next to it or you will be defending it in week 10.

Where the money goes in an IT project



Category	Typical items	Behaviour	Realtime Quiz
People	developers, testers, PM, contractors	60 to 80 % of most software projects; variable with duration	3,440 USD (96 %): the hours dominate
Licences	IDEs, Jira, design tools, per-seat SaaS	fixed per seat and month; renewals are easy to forget	0: IntelliJ Community, GitHub Free, draw.io, LibreOffice
Cloud and hosting	VPS, managed database, storage, bandwidth, TLS	variable with months and load; grows after go-live	Linode 4 GB 25 USD/month, domain 15 USD/year
Hardware	laptops, servers, network gear	capital cost, depreciated over years	students' own laptops: excluded, stated as assumption
Training	courses, books, onboarding time	often hidden inside "people"	Spring Boot 4 learning curve, carried as risk R1
Maintenance	bug fixes, upgrades, support, monitoring	15 to 20 % of build cost per year for years	≈ 80 h/year after the pilot, see TCO

A budget that shows only the build is half a budget. The Realtime Quiz cash spend is tiny (160 USD), but the *run* cost of 315 USD per year continues after the semester ends: someone must own it.

The cash budget: the part somebody actually pays

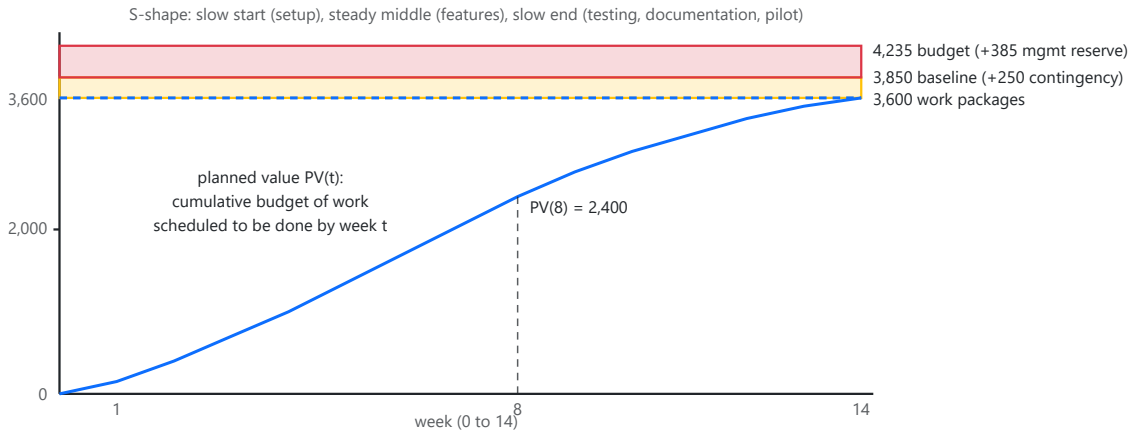
cash-budget.csv (non-labour costs, Realtime Quiz, 14 weeks)

Item	Unit price	Qty	Total	Note
Linode Shared 4 GB VPS	25 /month	4	100	weeks 1-14 = 4 bills
Domain gicitc.com	15 /year	1	15	renew every October
TLS certificate	0	1	0	Let's Encrypt via Caddy
Backups (Linode add-on)	5 /month	0	0	declined: nightly pg_dump
Paid tools (icon set, load test credit)	45 one-off	1	45	approved by sponsor
GitHub, draw.io, LibreOffice	0	-	0	free tiers
E-mail for notifications	0	-	0	free tier, 100 mails/day
Project cash total			160	
Run cost after go-live (per year)		12 × 25 + 15 = 315		

- Every line carries a **unit price, a quantity and a note** saying where the price came from (the vendor's price page and the date). Prices change; the note lets the next person re-check.
- Write the **zero lines**. "TLS 0" documents a decision (Let's Encrypt); leaving it out looks like you forgot it.
- Declined options ("backups add-on: declined") are the cheapest risk documentation you will ever write; they reappear in Lab-09 as risks.
- Separate **project cost** (until the pilot) from **run cost** (per year). The sponsor approves the first; the department must budget the second.

⚠ Cloud bills are variable: a 4 GB VPS fits the 2,000-connection NFR, but a resize to 8 GB doubles the monthly line. Carry that as a risk with a price tag (R4 on slide 13), not as a surprise.

Cost baseline: the time-phased budget you will be measured against



- **Cost estimate** (3,600): the roll-up of all work packages and purchases.
- **Contingency reserve** (250): money for *identified* risks ("known unknowns"). Owned by the project manager, spent through the change log.
- **Cost baseline** (3,850) = estimate + contingency, *time-phased*. This is the approved reference for earned value; changing it needs a formal change request.
- **Management reserve** (385): for *unidentified* risks ("unknown unknowns"). Owned by the sponsor, outside the baseline; releasing it moves money into the baseline.
- **Project budget** (4,235) = baseline + management reserve. Also called the funding requirement.

PMBOK Guide 7th ed.: the *performance measurement baseline* (PMB) integrates scope, schedule and cost baselines. Its cost total is the BAC used in earned value: here 3,600 for the work, with contingency tracked separately.

Sizing contingency: the risk-weighted (expected monetary value) approach

```
Contingency = Σ (probability × cost impact) over the identified risks

ID  Risk                                P     Impact (USD)         EMV
R1  Spring Boot 4 learning curve        50 %  40 h × 5   = 200     100
R2  WebSocket blocked by campus proxy   30 %  20 h × 5   = 100      30
R3  Load test fails, rework needed      40 %  30 h × 5   = 150      60
R4  VPS must grow to 8 GB (+25 USD/month)  40 %  2 mo × 25  =  50      20
R5  A student drops out, lecturer covers   20 %  10 h × 20  = 200      40
-------------------------------------------------------------------------
Contingency reserve                                                  250

Cost estimate           3,600
+ contingency         +   250   (6.9 % of the estimate)
= cost baseline         3,850
+ management reserve  +   385   (10 % of the baseline, sponsor's rule)
= project budget        4,235 USD

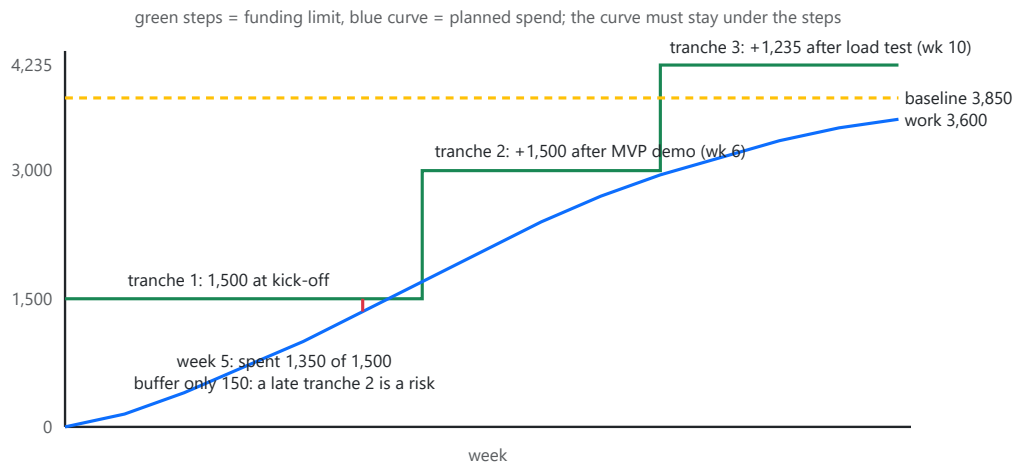
Spreadsheet: EMV in F2 is =D2*E2 ; reserve is =SUM(F2:F6)
```

Method	When	Realtime Quiz
Percentage of estimate	early, no risk list yet	10 % × 3,600 = 360
Risk-weighted (EMV)	risk register exists	250 (above)
Three-point spread	PERT estimates exist	Σ (E − M) × rate over branches, e.g. 8 h × 5 = 40 for WBS 4
Monte Carlo	large projects, tool support	P80 of the simulated total minus the estimate

- EMV is an *average*: if R1 happens it costs 200, not 100. The reserve works because not every risk fires; at 250 it covers any two of the five.
- Re-compute at every risk review: a closed risk releases its EMV back to the sponsor or to another risk.

⚠ Padding each activity secretly and adding a reserve on top is **double counting**. Estimates are honest most-likely values; risk money lives in the reserve, visibly.

Budget versus funding: the money must arrive before it is spent



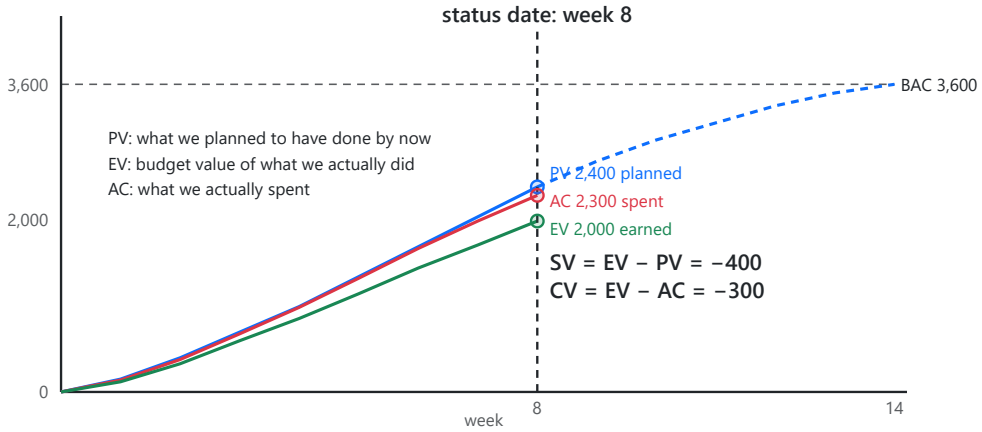
- **Budget** is what the work costs over time. **Funding** is when the sponsor makes cash available, usually in *steps* tied to gates (PRINCE2 7 calls them stage boundaries).
- **Cash flow** is the difference: negative cash flow (spend before funding) kills projects whose budget was perfectly correct.
- Funding steps are also a control: the sponsor can stop at a gate without losing the whole budget.

Cash flow by month (Realtime Quiz cash items only)

Month	Hosting	Domain	Tools	Out	Funded	Balance
Oct	25	15	45	85	160	+75
Nov	25	0	0	25	0	+50
Dec	25	0	0	25	0	+25
Jan	25	0	0	25	0	0

Labour is notional here; in a paid team the same table shows salaries leaving the account monthly.

Earned value: three curves answer "where are we, really?"



- **Planned value (PV)**: the budgeted cost of the work *scheduled* up to the status date, read from the time-phased baseline. Total PV at the end = **BAC**, budget at completion.
- **Earned value (EV)**: the budgeted cost of the work *actually completed*. Measured in budget units, not in hours spent: a work package budgeted at 480 USD that is done earns 480, however long it took.
- **Actual cost (AC)**: what the completed work *really cost*: hours × rate from timesheets plus invoices.
- Comparing spend with plan alone (AC vs PV: 2,300 vs 2,400, "we are under budget!") is the classic error. EV shows the team is **behind and over** at the same time.

Earned value comes from ANSI/EIA-748, summarised in the PMBOK Guide 7th ed. (Measurement performance domain). It needs three things you already have: a WBS, a time-phased baseline and a rule for measuring progress.

Variances and indices at week 8, every step shown

```
Status date: end of week 8            BAC = 3,600 USD
PV = 2,400     EV = 2,000     AC = 2,300

SV  = EV − PV = 2,000 − 2,400 = −400 USD   behind schedule
CV  = EV − AC = 2,000 − 2,300 = −300 USD   over budget
SPI = EV / PV = 2,000 / 2,400 = 0.833      83 % of planned pace
CPI = EV / AC = 2,000 / 2,300 = 0.870      0.87 USD of value per
                                           1 USD spent
% planned  = PV / BAC = 2,400 / 3,600 = 66.7 %
% complete = EV / BAC = 2,000 / 3,600 = 55.6 %
% spent    = AC / BAC = 2,300 / 3,600 = 63.9 %

In words: 56 % of the product exists, 64 % of the money is gone,
and 67 % should have been done. Both variances are negative.

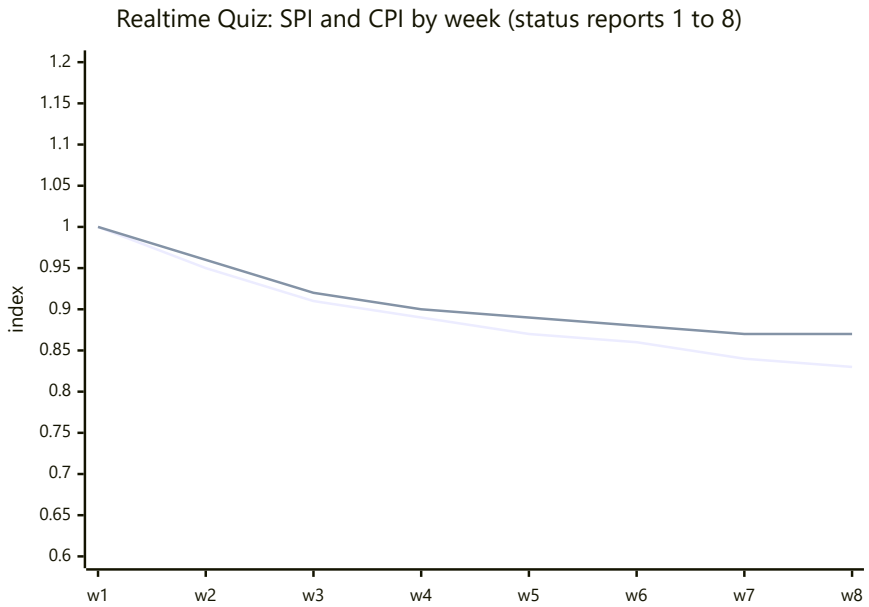
How EV was measured (per work package, from the WBS):
  1 Project management    LOE, earns as PV      400 of  700
  2 Setup and CI          done, 0/100           120 of  120
  3 Accounts              done                  240 of  240
  4 Quiz authoring        done                  480 of  480
  5 Live session core     50/50, started        300 of  600
  6 Scoring, leaderboard  % complete 75 %       210 of  280
  7 Analytics             2 of 4 milestones     140 of  240
  8-10 not started; 11 purchases paid so far    110 of  160
                                          EV = 2,000
```

Measure	Formula	Reads as	Week 8
Schedule variance	SV = EV − PV	negative = behind	−400
Cost variance	CV = EV − AC	negative = over budget	−300
Schedule performance index	SPI = EV / PV	< 1 behind, > 1 ahead	0.833
Cost performance index	CPI = EV / AC	< 1 over, > 1 under	0.870
Percent complete	EV / BAC	real progress	55.6 %

EV measurement rule	Use for
0/100: nothing until done	short packages (≤ 1 reporting period)
50/50: half at start, half at end	packages spanning two periods
Weighted milestones	long packages with checkable steps
Percent complete	only with an objective measure (tests passing, story points accepted)
Level of effort (LOE): EV = PV	management, stand-ups: never creates variance

⚠ Remember the mnemonic: EV comes **first** in every formula. Variances subtract, indices divide.

Indices over time: the trend matters more than one number



upper series at w8 = CPI 0.87, lower series = SPI 0.83. Source: `evm.csv`, one row per week, chart from the SPI and CPI columns.

Index	Green	Amber	Red
SPI, CPI	≥ 0.95	0.85 to 0.95	< 0.85
Action	report	analyse cause, plan recovery	escalate to sponsor, change request

- Thresholds belong in the **cost management plan** (Lab-06 Task 5), agreed before the first report, so nobody argues about them in week 8.
- Eight weeks of steady decline is not noise: the team is *systematically* slower than the estimate assumed (Spring Boot 4 learning curve, risk R1 came true).
- Research on large projects (Christensen, cited in the PMBOK Guide's EVM practice standard) finds that a CPI at 20 % completion rarely improves by more than 10 % afterwards. Plan with the trend, not with hope.

💡 Mermaid source is teaching material here: Lab-06 Task 4 draws exactly this chart from the weekly CSV.

Forecasting the end: estimate at completion by three formulas

```
Inputs (week 8): BAC 3,600  EV 2,000  AC 2,300  PV 2,400
                 CPI 0.8696  SPI 0.8333   remaining BAC − EV = 1,600

1  Variance was a one-off, rest goes to plan
   EAC = AC + (BAC − EV)
       = 2,300 + 1,600                    = 3,900 USD

2  Cost performance stays as it is (most used)
   EAC = BAC / CPI
       = 3,600 / 0.8696                   = 4,140 USD

3  Cost AND schedule performance both continue
   EAC = AC + (BAC − EV) / (CPI × SPI)
       = 2,300 + 1,600 / (0.8696 × 0.8333)
       = 2,300 + 1,600 / 0.7246
       = 2,300 + 2,208                    = 4,508 USD

4  Bottom-up re-estimate of the remaining work (team, week 8)
   ETC = 340 h × 5 = 1,700 USD  →  EAC = 2,300 + 1,700 = 4,000 USD

Spread of the forecast: 3,900 .. 4,508. Report the range and say
which assumption you believe; for Realtime Quiz the trend on the
dashboard argues for formula 2 or 3, not for 1.
```

Formula	Assumption	Use when	Week 8
EAC = AC + (BAC − EV)	future work at budgeted rate	the cause is fixed (one bad week, a one-off purchase)	3,900
EAC = BAC / CPI	cost efficiency continues	default; the cause is systemic (rates, productivity)	4,140
EAC = AC + (BAC − EV) / (CPI × SPI)	schedule pressure also raises cost	a fixed deadline will force overtime or extra people	4,508
EAC = AC + bottom-up ETC	team re-estimates the rest	original estimate no longer valid; expensive but best	4,000

- **ETC** (estimate to complete) = EAC − AC: how much more money is needed from today. Formula 2: 4,140 − 2,300 = 1,840.
- **VAC** (variance at completion) = BAC − EAC: 3,600 − 4,140 = −540 USD, the expected overrun.
- Against the baseline of 3,850 (with contingency), formula 2 still overruns by 290: a request for management reserve, with the sponsor.

⚠ Formula 1 is the optimist's favourite. Use it only when you can name the one-off cause and show it is closed.

TCPI: is the target still achievable, and what to do if not

```
TCPI = efficiency needed on the REMAINING work to hit a target

To finish within BAC (3,600):
  TCPI = (BAC − EV) / (BAC − AC)
       = (3,600 − 2,000) / (3,600 − 2,300)
       = 1,600 / 1,300                      = 1.231

  Meaning: every remaining dollar must earn 1.23 of value, while
  the team has earned 0.87 so far. A 42 % jump in efficiency:
  not credible → BAC is lost, re-baseline.

To finish within EAC (formula 2, 4,140):
  TCPI = (BAC − EV) / (EAC − AC)
       = 1,600 / (4,140 − 2,300) = 1,600 / 1,840   = 0.870

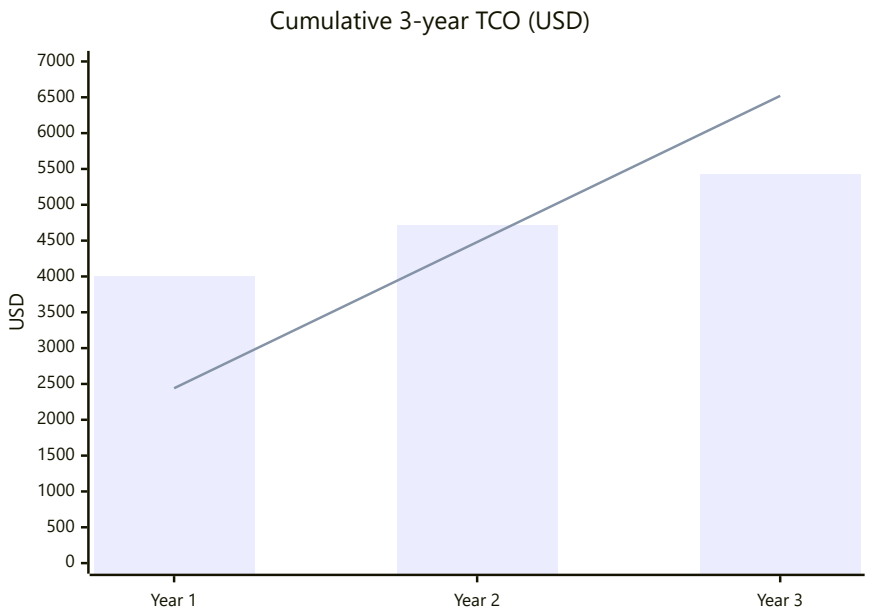
  Meaning: equals today's CPI → achievable if nothing changes.
  Ask the sponsor for 4,140, not 3,600.

Rule of thumb: TCPI more than 0.10 above CPI is a red flag.
```

Option when EAC > budget	Effect on cost	Realtime Quiz (EAC 4,140, budget 3,850)
Reduce scope (drop "Could" features)	lowers BAC and EAC	drop chat and guest join: −40 h = −200 → EAC 3,940
Improve efficiency (pairing, remove blockers, reuse)	raises CPI on remaining work	reuse Lab-03 code for export: −20 h = −100
Crash (add lecturer hours to critical path)	saves time, raises cost	+10 h × 20 = +200 to hold the week 14 pilot
Use contingency (R1 fired)	within baseline	release R1's 100 of the 250
Ask for management reserve and re-baseline	new BAC via change request	request 290 of the 385; new baseline 4,140
Accept the overrun and report it	none, transparent	only with sponsor agreement in writing

💡 The combination usually wins: drop chat (−200), use R1 contingency (100), request 90 of reserve. Chapter 02's trade-offs (scope, time, cost) return here with numbers attached.

Total cost of ownership: build and self-host versus a SaaS quiz tool



bars = build and self-host (cumulative), line = SaaS subscription (cumulative). The lines cross during year 3.

Cost line	Year 1	Year 2	Year 3	3-year
Option A: build (Realtime Quiz) and self-host				
Development (estimate, incl. 4 months hosting)	3,600	0	0	3,600
Hosting 25/month, remaining months, + domain 15	200	315	315	830
Maintenance 40 h then 80 h/year × 5	200	400	400	1,000
Total A	**4,000**	**715**	**715**	**5,430**
Option B: SaaS quiz tool, 10 instructor seats				
Subscription 10 × 17 USD × 12	2,040	2,040	2,040	6,120
Onboarding and training, lecturer 20 h × 20	400	0	0	400
Total B	**2,440**	**2,040**	**2,040**	**6,520**

- **TCO** adds acquisition, operation, support and retirement over a stated horizon (3 to 5 years for software). Cheapest year 1 is rarely cheapest overall.
- Include what the option makes you pay elsewhere: the SaaS needs no server but still needs training; the build needs a maintainer after the students graduate.

ROI, payback and NPV of building instead of subscribing

```
Benefit of building = SaaS cost avoided over 3 years = 6,520 USD
Cost of building    = 3-year TCO of option A         = 5,430 USD

ROI = (benefit − cost) / cost
    = (6,520 − 5,430) / 5,430 = 1,090 / 5,430        = 20.1 %

Payback: when does cumulative A drop below cumulative B?
  end of Y1   A 4,000  B 2,440   A is 1,560 behind
  end of Y2   A 4,715  B 4,480   A is   235 behind
  during Y3   A gains (170 − 59.6) = 110.4 per month
              235 / 110.4 = 2.1 months → payback ≈ 26 months

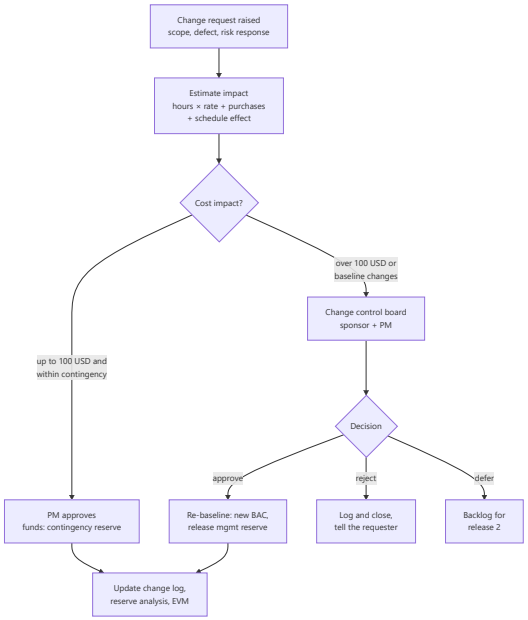
NPV at a 10 % discount rate (net cash flow B − A per year):
  Y1  2,440 − 4,000 = −1,560   / 1.10  = −1,418
  Y2  2,040 −   715 = +1,325   / 1.21  = +1,095
  Y3  2,040 −   715 = +1,325   / 1.331 = +  996
  NPV                                  = +  673 USD  (positive)
```

Measure	Formula	Reads as	Here
ROI	(benefit − cost) / cost	return per dollar over the horizon	20 %
Payback period	time until cumulative benefit = cost	how long the money is at risk	26 months
NPV	Σ cash flow$_t$ / $(1 + r)^t$	value today; positive = worth doing	+673
Benefit-cost ratio	benefit / cost	> 1 = benefits exceed costs	1.20

- A 20 % ROI over three years with a 26-month payback is a **weak financial case**: the decision rests on the non-financial benefits (own data, CSV/PDF export, no seat limit, the capstone's learning outcome). Write them in the business case as benefits, and say which are measurable.
- Sensitivity: if the SaaS costs 12 instead of 17 per seat, B totals 4,720 and building never pays back. Always test the one or two inputs that drive the result.
- PRINCE2 7 keeps the business case alive: re-check ROI at every stage boundary, not only at approval.

⚠ Do not count the students' notional hours as "free". A cost that nobody invoices is still a cost the department bears in time.

Cost control: every change gets a price before it gets a yes




```
CR-07  Bundle Noto Sans Khmer font in the PDF export
Raised by: tester, week 9 (Khmer names render as boxes)
Impact estimate:
  research 2 h + code 6 h + tests 4 h = 12 h × 5 USD =  60 USD
  schedule: +0 days (parallel to WBS 8, not on critical path)
Source of funds: contingency reserve (risk "PDF fonts", Lab-09)
Decision: PM approves (≤ 100 USD rule), logged as CR-07
Baseline: unchanged (BAC stays 3,600)
Reserve analysis after CR-07:
  contingency 250 − 100 (R1 fired) − 60 (CR-07) = 90 USD left
  management reserve 385 untouched
```

- **Thresholds** (who may approve what) live in the cost management plan; the flow above uses 100 USD.
- Approved changes update **the baseline** only through the CCB; contingency spending changes the *reserve*, not the BAC.
- Rejected and deferred requests are still logged: they are evidence when the sponsor asks "why did it cost more?"

The weekly cost status report: one page the sponsor can read in two minutes

```
# Cost status · Realtime Quiz · week 8 of 14 · status date 2026-11-22

| Measure | Value | Threshold | Status |
|---|---|---|---|
| BAC / cost baseline / budget | 3,600 / 3,850 / 4,235 | | |
| PV · EV · AC | 2,400 · 2,000 · 2,300 | | |
| SV · CV | −400 · −300 | | red |
| SPI · CPI | 0.83 · 0.87 | amber 0.85–0.95 | red · amber |
| EAC (BAC/CPI) · VAC | 4,140 · −540 | EAC ≤ 3,850 | red |
| TCPI to BAC | 1.23 | ≤ CPI + 0.10 | red |
| Contingency used / left | 160 / 90 | | amber |
| Management reserve | 385 / 385 | | green |

## Variance analysis
Root cause: WBS 5 Live session took 96 h against 70 h estimated
(Spring Boot 4 WebSocket configuration, risk R1). Scoring is on
plan; analytics started one week late.

## Forecast and recommendation
EAC range 3,900–4,508, most likely 4,140. Recommend: drop chat
(−200), release R1 contingency (100), request 90 of management
reserve. Decision needed by the CCB on 2026-11-25.
```

- **Same numbers every week**, same order, thresholds printed next to them: the reader learns the layout once.
- **Variance analysis** names the work package and the cause, not "the team was slow".
- **Reserve analysis** shows how much cushion is left; a shrinking contingency with six weeks to go is itself a warning.
- **Forecast plus recommendation**: a report without a proposed decision just transfers worry.
- Control loop: measure (timesheets, invoices) → compare (EVM) → forecast (EAC) → act (change request or recovery) → update the plan. PMBOK Guide 7th ed. calls this the Measurement performance domain.

> 💡 Lab-06 Task 4 asks for this report at two status dates, and the Challenge turns the week-9 version into a sponsor briefing with a chart.

WRAP-UP

Best practices for managing IT project costs

1 Estimate twice, by different routes

Parametric (points × rate) and bottom-up (WBS roll-up) should agree within 20 %. When they do not, the disagreement is the most valuable finding of the planning phase.

2 Write the basis of estimate

Rate, productivity, exclusions, accuracy class, date of the vendor price. Zero lines ("TLS: 0, Let's Encrypt") document decisions.

3 Keep reserves visible and separate

Honest most-likely estimates, a contingency sized from the risk register (EMV), a management reserve owned by the sponsor. No padding inside activities.

4 Time-phase the baseline in a spreadsheet

One row per activity, one column per week, `=SUM()` per week, a cumulative row: that is PV(t). Without it there is no earned value.

5 Measure EV with objective rules

0/100 or 50/50 for short packages, accepted story points or passing tests for percent complete. "About 80 % done" is not a measurement.

6 Forecast as a range and recommend

Report EAC by two or three formulas, say which you believe and why, and attach the recovery options with their prices. Decisions, not just data.

Common mistakes (and the slide that prevents each one)

Comparing spend with plan

"AC 2,300 under PV 2,400, we are fine." Without EV you cannot tell under-spending from being behind. Slide 15.

The ROM becomes the promise

A week-0 "about 2,000" quoted without its −50/+100 % range is defended in week 10 as if it were definitive. Slide 9.

Sunk cost reasoning

"We already spent 1,200, we must finish." Compare cost-to-complete against value, from today. Slide 5.

Budgeting only the build

Hosting, domain renewal, maintenance and training continue after go-live; a TCO horizon of 3 years shows them. Slides 10, 20.

Optimistic EAC

Choosing EAC = AC + (BAC − EV) after eight weeks of falling CPI. Use BAC / CPI unless you can name and close the one-off cause. Slide 18.

Silent scope changes

A "small" font fix costs 60 USD; twenty of them cost the contingency. Every change gets an estimate and a log entry. Slide 22.

Chapter quiz 10 questions

1. At the end of week 6 the team has spent 1,200 USD and considers switching to a SaaS tool. Which amount must NOT influence the decision?

- The 2,250 USD still needed to finish building
- The 460 USD the SaaS route costs from here
- The 1,200 USD already spent
- The benefit of owning the data and exports

2. Student hours charged at 5 USD/h that grow with every extra week of work are which cost type?

- Direct and variable
- Direct and fixed
- Indirect and variable
- Indirect and fixed

3. Parametric estimate: the backlog has 120 story points, the historical rate is 6 hours per point and the rate is 5 USD/h. Labour cost = ?

- 720 USD
- 3,000 USD
- 3,600 USD
- 4,320 USD

4. Three-point estimate for Quiz authoring: O = 80 h, M = 96 h, P = 160 h. The PERT expected effort is:

- 96 h
- 104 h
- 112 h
- 120 h

5. A rough order of magnitude estimate is given for a 3,600 USD project. Which range fits the ROM class used in this chapter?

- 3,420 to 3,960 USD
- 3,060 to 4,680 USD
- 1,800 to 7,200 USD
- 3,600 USD exactly

6. Which statement about reserves is correct?

- The management reserve is part of the cost baseline and is spent by the project manager
- The contingency reserve covers identified risks and is inside the cost baseline; the management reserve covers unknown unknowns and is owned by the sponsor
- Contingency is always 10 % of the estimate
- Reserves are hidden inside each activity estimate

7. At week 8: PV = 2,400, EV = 2,000, AC = 2,300. SPI = ?

- 0.83
- 0.87
- 1.15
- 1.20

8. With BAC = 3,600, EV = 2,000 and AC = 2,300, the EAC assuming the current cost performance continues is closest to:

- 3,900 USD
- 4,140 USD
- 4,508 USD
- 3,600 USD

9. Building costs 5,430 USD over three years and avoids a SaaS bill of 6,520 USD. The ROI is about:

- −17 %
- 17 %
- 20 %
- 120 %

10. A change request needs 12 extra hours at 5 USD/h and the cost management plan lets the PM approve changes up to 100 USD from contingency. What happens?

- The PM approves the 60 USD change from contingency and logs it; the baseline is unchanged
- The BAC rises by 60 USD automatically
- The sponsor must release management reserve
- The change is rejected because contingency is for unknown risks

WRAP-UP

Summary

Classify every cost twice: direct or indirect, fixed or variable. Sunk costs never decide; opportunity costs often should.

Estimate by two routes: parametric (120 points × 6 h × 5 USD = 3,600) and bottom-up (WBS roll-up 3,600); three-point gives the range.

State the accuracy class with every number: ROM −50/+100 %, budgetary −15/+30 %, definitive −5/+10 %.

IT budgets are people plus licences, hosting, hardware, training and years of maintenance; budget the run cost, not only the build.

Estimate 3,600 + contingency 250 (EMV of the risks) = cost baseline 3,850; + management reserve 385 = budget 4,235. Funding arrives in steps; watch the cash flow.

Earned value: PV planned, EV earned, AC spent. SV = EV − PV, CV = EV − AC, SPI = EV / PV, CPI = EV / AC. Week 8: 0.83 and 0.87.

Forecast with EAC = BAC / CPI (4,140) as default, show the 3,900 to 4,508 range, and test feasibility with TCPI (1.23 to BAC: not credible).

TCO over 3 years and ROI, payback and NPV decide build versus SaaS; a 20 % ROI with 26-month payback needs non-financial benefits to win.

Cost control: every change request gets a price, a funding source and a log entry; the baseline changes only through the CCB.

Open Lab-06: 5 tasks + 1 challenge

Next chapter: 07 · Quality Management